

Enterprise Information Systems

PRESENTATION

Core Problem

£200M annual loss from food waste and stockouts.

Manual stock counts causing delays.

Fragmented systems across 4,000+ stores.

Legacy infrastructure limiting scalability and accuracy.

Key Insights from Analysis

Inventory data lacks real-time accuracy.

Major gaps between B2C and B2B ecosystems.

BI systems struggle with latency and governance.

GDPR compliance weaknesses in consent, retention, and data flows.

Ethical risks: job displacement, supplier inequality, increased surveillance.

IS Strategy Overview

A unified digital ecosystem enabling:

Real-time inventory visibility.

Automated replenishment.

Integrated supplier collaboration.

Advanced BI and predictive analytics.

GDPR-compliant data lifecycle.

Infrastructure Modernization

Target Architecture:

70% cloud migration via Azure.

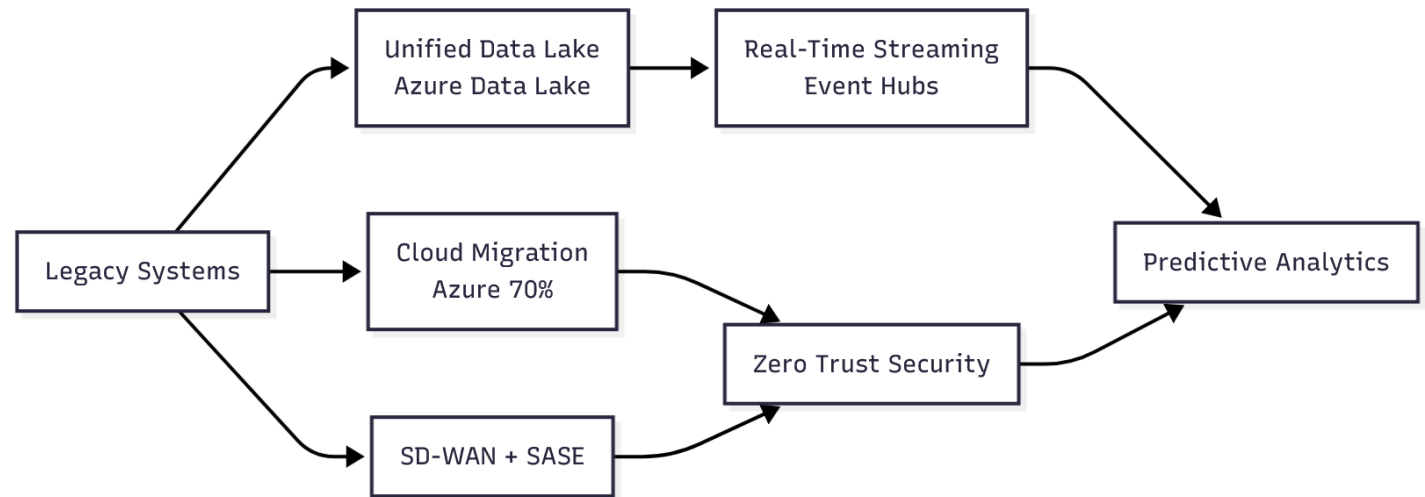
SD-WAN and SASE for secure connectivity.

Unified data lake + real-time streaming.

Zero Trust security model. Expected Impact:

40% lower infrastructure cost.

99.99% availability



Slide 7: Business Intelligence Transformation

Problems Today:

Slow batch ETL.

Siloed data definitions.

Limited unstructured data processing. **Solution:**

Real-time analytics.

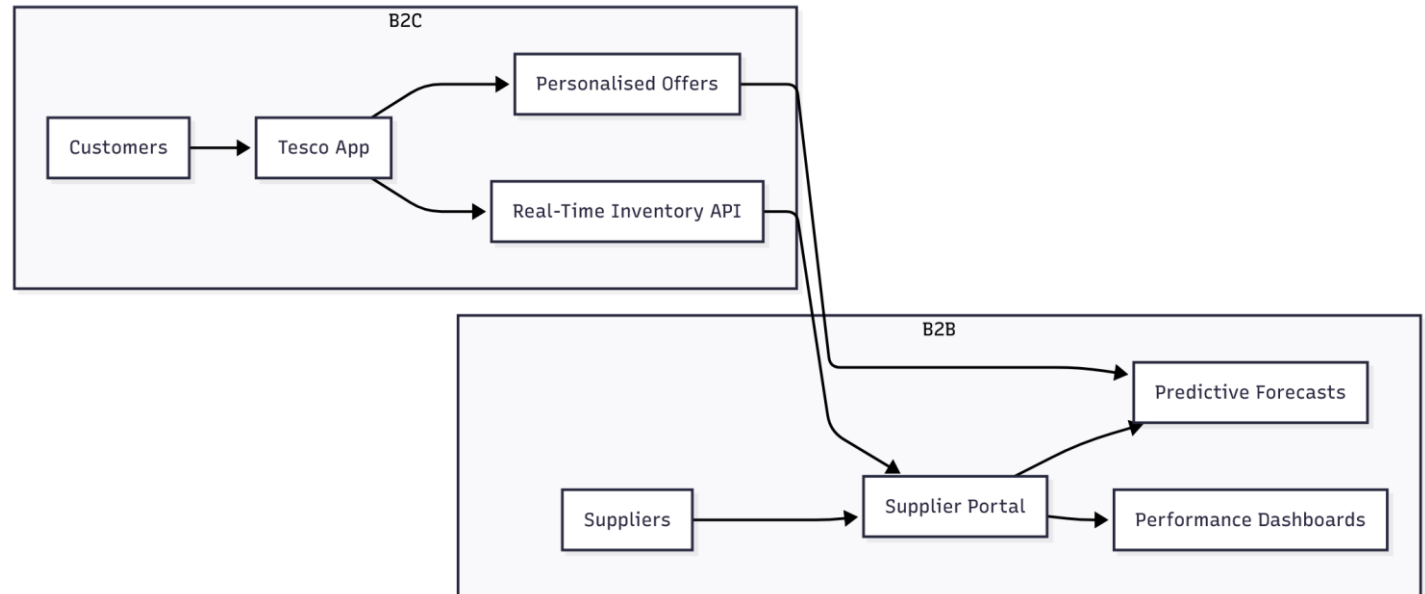
Predictive demand forecasting.

Personalized pricing.

Outcomes:

£45M saved in waste.

30% less manual reporting.



B2C/B2B Ecosystem Integration

Current Gaps:

Supplier data lag: 4.2 hours.

High cart abandonment (68%). **Proposed Fixes:**

API-first integration.

Real-time supplier dashboards.

Personalized mobile experience tied to live store inventory.

Benefits:

35% lower cart abandonment.

28% better supplier on-time delivery.

GDPR & Privacy Compliance

Current Risks:

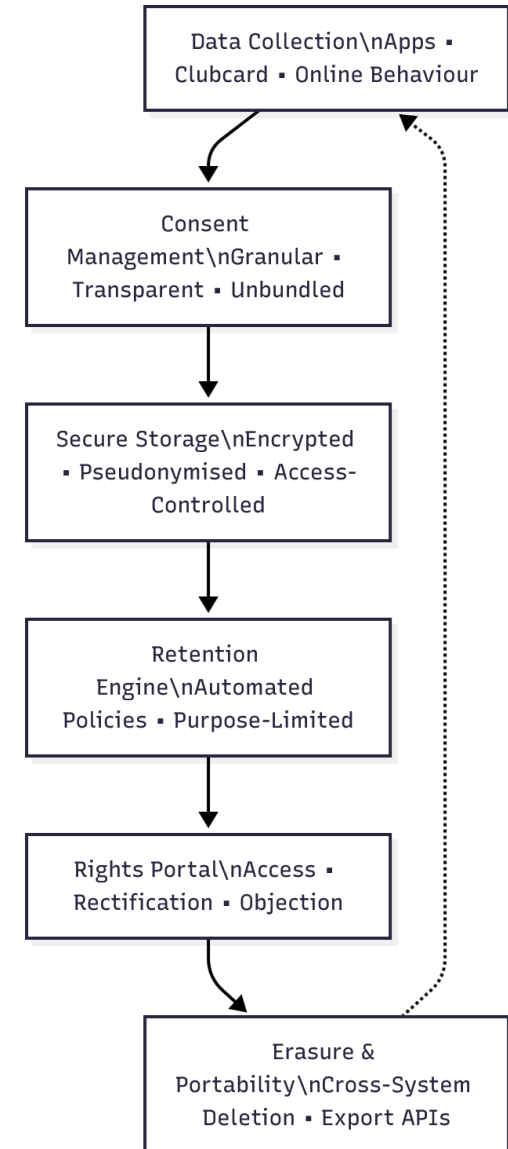
- Bundled/unclear consent.
- Indefinite data retention.
- Weak pseudonymisation.
- Slow subject access responses.

Strategy:

- Layered consent model.
- Automated retention rules.
- PIAs mandatory for new systems.
- Cross-system erasure workflow.

Result:

- Better trust, compliance, and lower breach risks.



Ethics & Sustainability

Job anxiety among staff.

Supplier disadvantages.

Increased energy consumption.

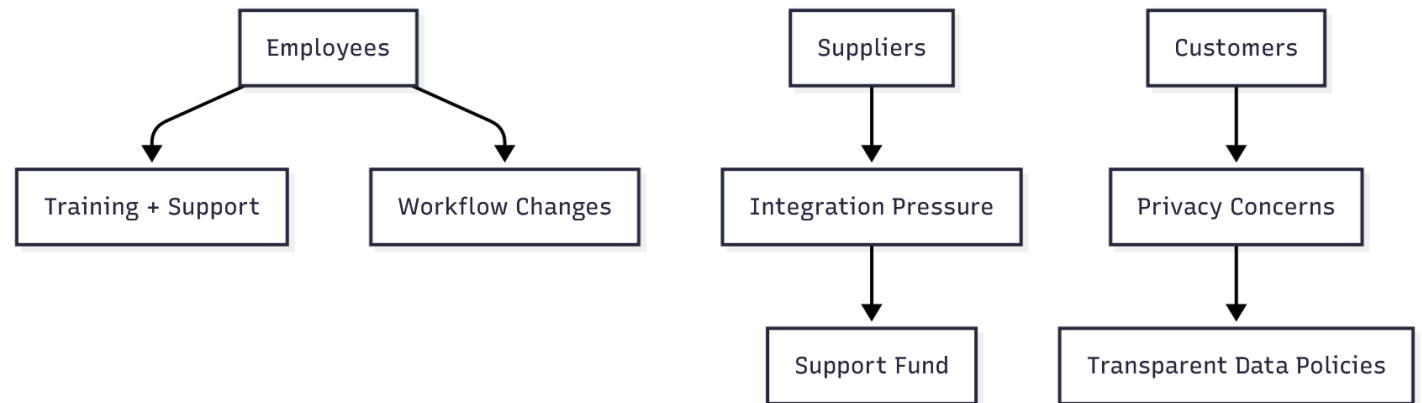
Mitigations:

Staff retraining programs.

Supplier support fund.

Reuse/recycling of devices.

Transparent data practices.



Business Model Transformation

Before IS:

Manual, reactive, siloed.

After IS:

Real-time inventory.

Automated forecasting.

Supplier collaboration.

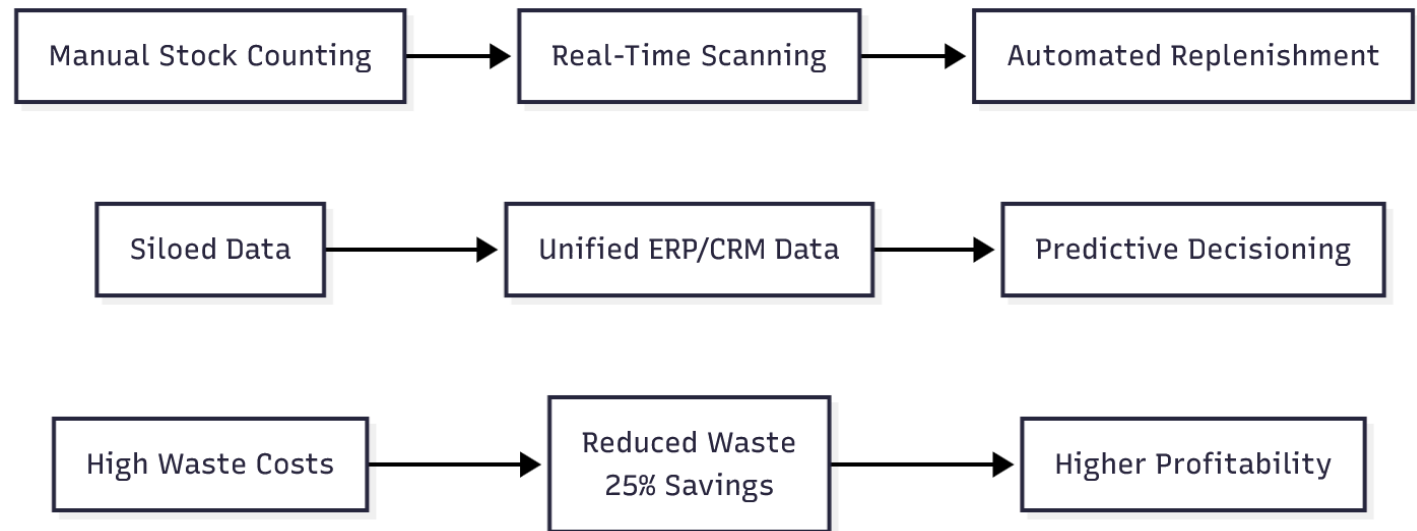
Data-driven customer experience.

New Value:

Better availability.

Lower waste.

New B2B data revenue.



Real-World IT Trends Alignment

Cloud migration & edge computing.

AI-driven personalization.

IoT for live stock monitoring.

Zero Trust cybersecurity.

Privacy-by-design frameworks.

Implementation Roadmap

Phase 1 (1–6 months): Cloud foundations, data mapping, security.

Phase 2 (7–12 months): API rollout, BI modernization, supplier onboarding.

Phase 3 (13–18 months): Real-time analytics, AI forecasting, optimization.

Conclusion – Strategic Impact

25% waste reduction.

Higher availability and customer satisfaction.

Stronger supplier collaboration.

GDPR-aligned, ethical digitalization.

Tesco gains durable competitive advantage through integrated IS strategy.